

SKILLS NEEDED TO BECOME A DATA SCIENTIST

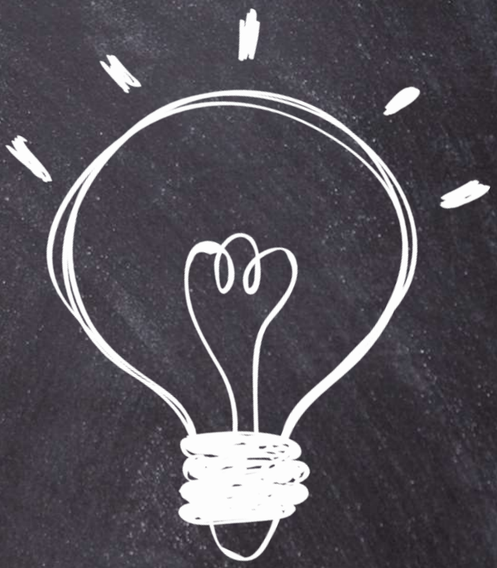




INTRODUCTION TO DATA SCIENCE

- WHAT IS DATA SCIENCE?

A multidisciplinary field that combines statistics, computer science, and domain knowledge to extract insights from data.



- Importance of Data Science:
- Driving decisions with data.
- Powering innovation in industries like healthcare, finance, and technology.

CORE TECHNICAL SKILLS

PROGRAMMING LANGUAGES:

- Python: Data manipulation, analysis, and visualization.
- R: Statistical analysis and modeling.

Data Wrangling & Manipulation Tools:

- SQL: Managing and querying databases.
- Pandas & NumPy: Data preprocessing in Python.

STATISTICAL AND ANALYTICAL SKILLS

Understanding of Statistics:

- Descriptive statistics, probability, hypothesis testing.

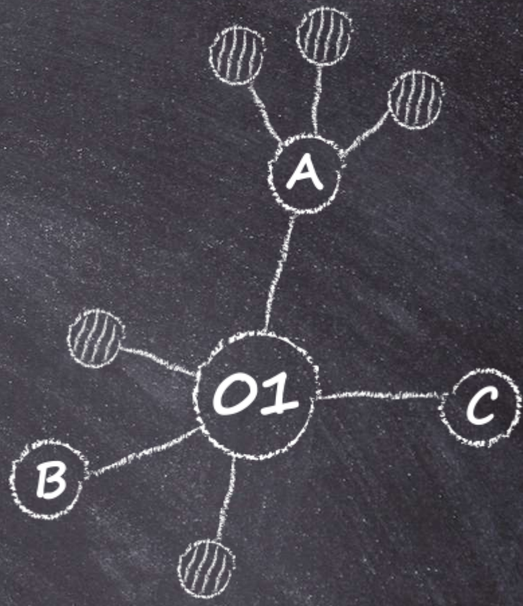
Analytical Mindset:

- Drawing actionable insights from data trends.

Tools:

- Excel, SPSS, SAS.

MACHINE LEARNING SKILLS



KEY CONCEPTS:

- Supervised and unsupervised learning.
- Feature engineering and model evaluation.

Tools & Frameworks:

- Scikit-learn, TensorFlow, PyTorch.

Practical Application:

- Building predictive models to solve business problems.

DATA VISUALIZATION AND COMMUNICATION

IMPORTANCE OF VISUALIZATION:

- Transforming raw data into understandable insights.

Popular Tools:

- Tableau, Power BI, Matplotlib, Seaborn.

Soft Skills:

- Presenting insights effectively to stakeholders.



BIG DATA AND CLOUD COMPUTING SKILLS

BIG DATA TOOLS:

- Hadoop, Spark for processing massive datasets.

CLOUD PLATFORMS:

- AWS, AZURE, GOOGLE CLOUD FOR SCALABLE STORAGE AND COMPUTING.

RELEVANCE:

- MANAGING LARGE-SCALE DATA INFRASTRUCTURE.



PROBLEM-SOLVING AND DOMAIN EXPERTISE

CRITICAL THINKING:

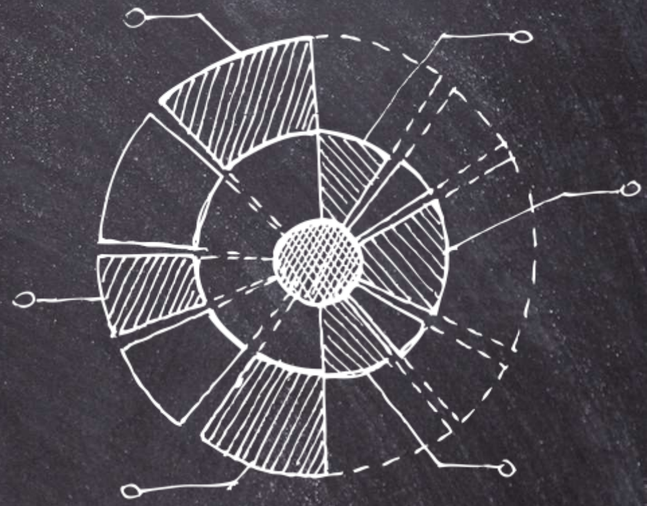
- Solving complex business challenges through data.

Domain Knowledge:

- Understanding the industry you're working in (e.g., healthcare, finance).

Real-World Application:

- Collaborating with teams to create impact.



CONCLUSION

DATA SCIENCE DEMANDS A MIX OF TECHNICAL SKILLS, ANALYTICAL THINKING, AND EFFECTIVE COMMUNICATION. BY MASTERING KEY AREAS LIKE PROGRAMMING, STATISTICS, AND MACHINE LEARNING, YOU CAN TACKLE COMPLEX PROBLEMS AND MAKE DATA-DRIVEN DECISIONS. KEEP LEARNING AND GAINING EXPERIENCE TO EXCEL IN THIS DYNAMIC FIELD!

THANK YOU